



ANGLO-CHINESE JUNIOR COLLEGE
Preliminary Examination 2011

BIOLOGY

8875/01

HIGHER 1

26 August 2011

Paper 1 Multiple Choice

1 hour

Additional Materials: Multiple Choice Answer Sheet

READ THESE INSTRUCTIONS FIRST

Write in soft pencil.

Do not use staples, pencil clips, highlighters, glue or correction fluid.

Write your name, Centre number and index number on the Answer Sheet provided.

There are **thirty** questions in this paper. Answer **all** questions. For each question there are four possible answers, **A, B, C** and **D**.

Choose the **one** you consider correct and record your choice in **soft pencil** on the separate answer sheet.

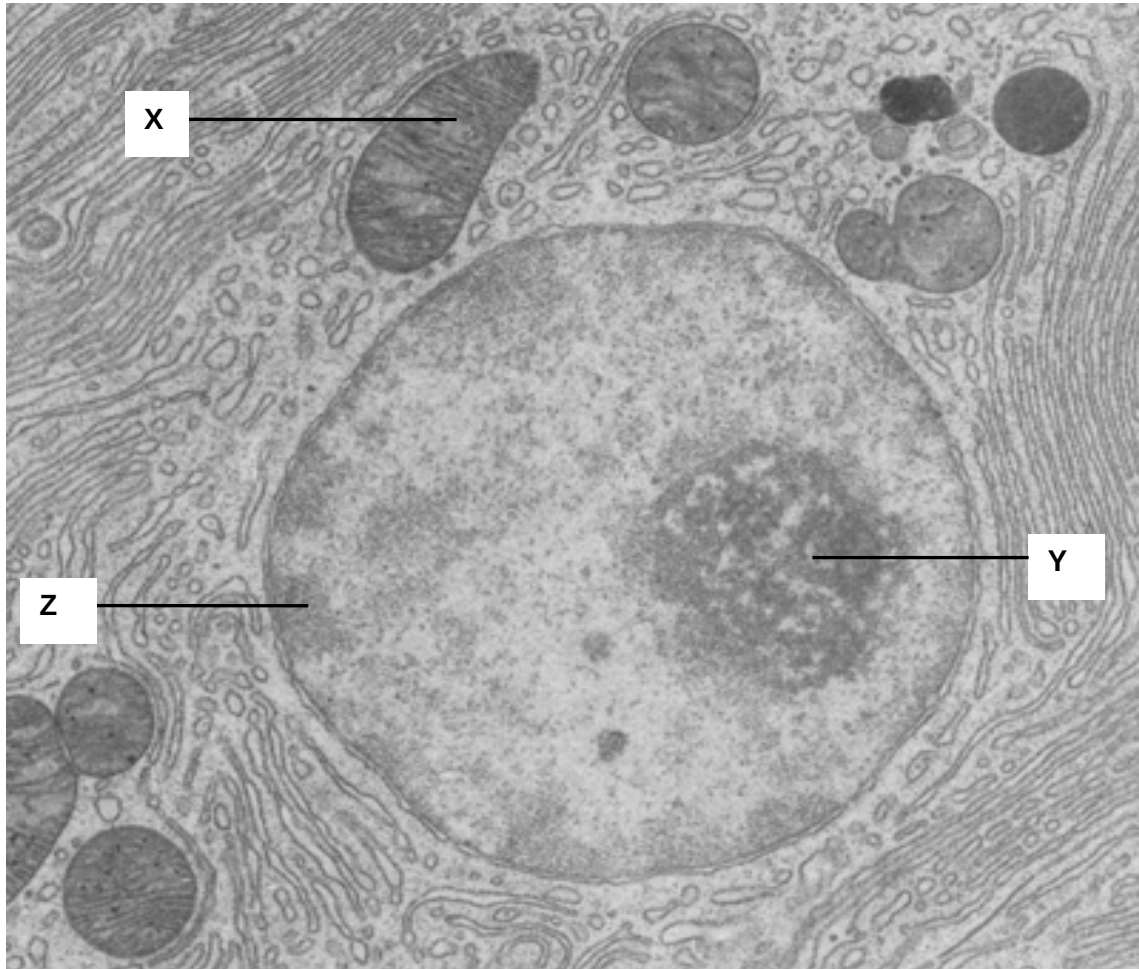
Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.
Any rough working should be done in this booklet.

Calculators may be used.

This document consists of **16** printed pages.

- 1 The electronmicrograph below shows the structures found in a cell.



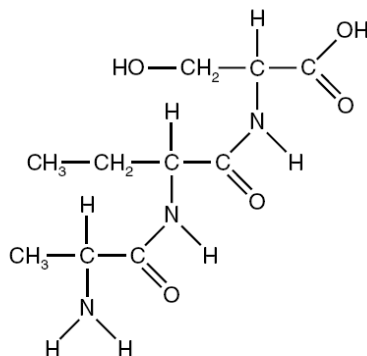
Which of the following descriptions are true about structures X, Y and Z?

	X	Y	Z
A	Contains reduced NAD^+ and reduced FAD	Transcription of gene coding for ribosomal RNA	Contains heterochromatin which is transcriptionally inactive
B	Involved in oxidative phosphorylation	Transcription of gene coding for ribosomal protein	Contains heterochromatin which is transcriptionally active
C	Contains reduced NAD^+ and reduced FAD	Involved in assembly of ribosomal subunits	Contains euchromatin which is transcriptionally inactive
D	Involved in oxidative phosphorylation	Involved in synthesis of ribosomal subunits	Contains euchromatin which is transcriptionally active

- 2 Which row best describes the primary roles of membranes found in the respective organelles?

	Mitochondrial inner membrane	Lysosomal membrane
A	Increases surface area for ATP synthesis	Maintains low pH with the aid of protein channels for the activation of enzymes
B	Increases surface area for ATP hydrolysis	Maintains neutral pH with the aid of protein pumps for the inactivation of enzymes
C	Increases surface area for ATP hydrolysis	Maintains low pH with the aid of protein channels for the inactivation of enzymes
D	Increases surface area for ATP synthesis	Maintains low pH with the aid of protein pumps for the activation of enzymes

- 3 The diagram below shows a biological molecule found in living cells.



- 1 It is soluble in water.
- 2 Double bond between the carbon and oxygen atoms increases membrane fluidity.
- 3 It can act as a buffer against extreme pH change.
- 4 It can undergo condensation to form a polymer.

Which of the following statements are true about this molecule?

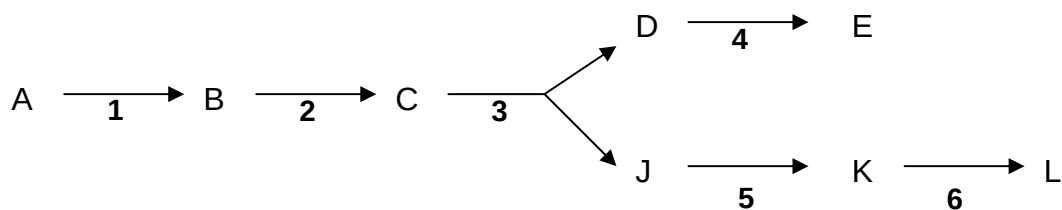
- A 1 and 3
 B 2 and 4
 C 2 and 3
 D 1 and 4

4 With reference to the descriptions below, identify the molecules **P, Q, R, S** and **U**.

- | | | |
|---|---|----------|
| 1 | (a) Soluble in water..... | 2 |
| | (b) Insoluble in water..... | 3 |
| 2 | (a) Red precipitate is observed in Benedict's solution..... | P |
| | (b) Benedict's solution remains blue | Q |
| 3 | (a) Cross links are present..... | R |
| | (b) Cross links are absent..... | 4 |
| 4 | (a) Lower calorific value per unit volume..... | S |
| | (b) Higher calorific value per unit volume..... | U |

	P	Q	R	S	U
A	Fructose	Sucrose	Cellulose	Glycogen	Triglyceride
B	Sucrose	Fructose	Amylopectin	Amylose	Glycogen
C	Lactose	Sucrose	Glycogen	Amylose	Triglyceride
D	Maltose	Fructose	Cellulose	Amylopectin	Glycogen

5 In the series of metabolic reactions shown below, enzyme 3 catalyzes the break down of C to D and J. Product E, a substrate for another reaction, inhibits enzyme 3.



Which of the following would likely to happen if product E accumulates?

- A** The rate of production of B will decrease.
- B** The rate of production of D and E will increase.
- C** The rate of production of J, K and L will increase.
- D** The rate of production of D, E, J, K and L will decrease.

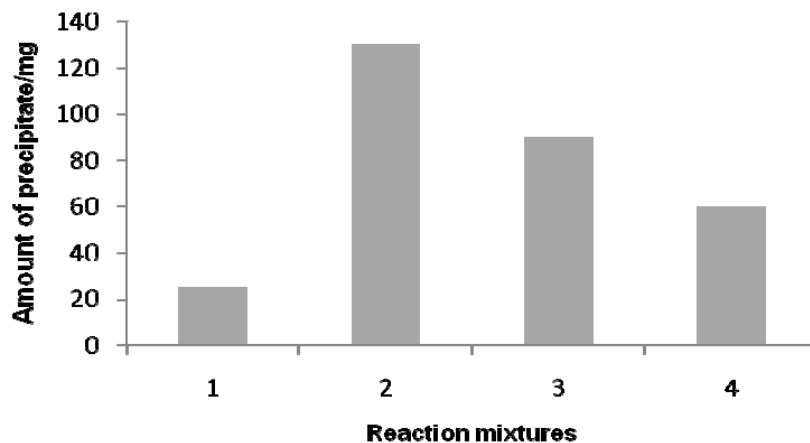
- 6 Creatine kinase is active in the intermembrane space of mitochondria. The enzyme activity shows a reading of 50 arbitrary units (A.U.) under normal cellular condition. The enzyme activity of creatine kinase was also measured after incubation in the following solutions:

- 1 Deionized water.
- 2 Acetic acid.
- 3 Sodium hydroxide.

Which of the following findings are consistent with the expected results?

	1	2	3
A	30 A.U.	50 A.U.	10 A.U.
B	50 A.U.	30 A.U.	10 A.U.
C	30 A.U.	10 A.U.	50 A.U.
D	10 A.U.	30 A.U.	50 A.U.

- 7 Benedict's Test is often used to determine the presence and relative quantity of reducing sugars in samples. In the presence of reducing sugars, red copper oxide precipitate is obtained. The higher the concentration of reducing sugars, the greater the amount of red precipitate obtained. The graph below shows the amount of precipitate measured in reaction mixtures under different experimental conditions.

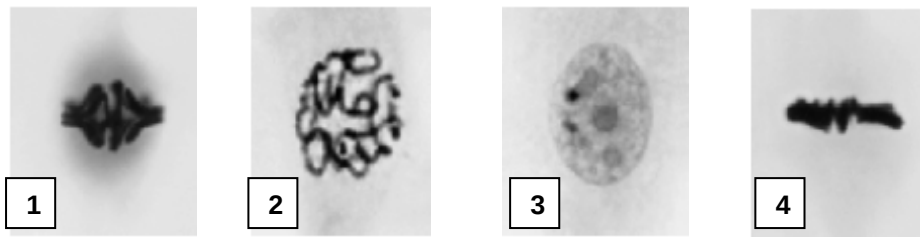


- 1 2% enzyme X + 10% starch + 28°C
- 2 2% enzyme X + 10% glycogen + 2% HCl + 100°C
- 3 2% enzyme X + 10% amylose + 2% HCl + 100°C
- 4 2% enzyme X + 10% starch + 60°C

Based on the data shown above, which conclusion is valid?

- A Change in pH is essential to the enzyme activity.
- B Rate of reaction is not affected by the temperature.
- C The degree of branching in substrate molecule is inversely proportional to the rate of reaction.
- D Many cysteine residues are present in the enzyme.

- 8 The photomicrographs below depict the different stages of a typical mammalian cell cycle.



Which of the above photomicrograph(s) could be obtained from the following cells?

- | | Lymphocyte | Erythrocyte (red blood cell) |
|----------|-------------------|-------------------------------------|
| A | All of the above | 3 only |
| B | 1, 2 and 4 | None of the above |
| C | All of the above | None of the above |
| D | 3 only | 1, 2 and 4 |
- 9 Meiosis is known to produce four daughter cells containing half the original number of chromosomes in the parent cell.

The table below shows the number of chromosomes in each gamete after meiosis has taken place in human germ cells.

Gametes 1-4 were derived from germ cell X, whereas gametes 5-8 were derived from germ cell Y.

Germ cell X		Germ cell Y	
Gametes	Number of chromosomes	Gametes	Number of chromosomes
1	22	5	23
2	24	6	23
3	24	7	22
4	22	8	24

Which of the following description is consistent with the data shown in the table above?

- A** Homologous chromosomes could not pair up to form bivalents in germ cell X.
- B** A pair of chromatids could not separate during anaphase II in germ cell X.
- C** Centromere of a pair of chromatids failed to divide during anaphase II in germ cell Y.
- D** Homologous chromosomes could not separate during anaphase I in germ cell Y.

- 10** The following coding sequence of DNA is taken from a random eukaryote gene.

3' TTACGCTTCGAAATAGGAATATCATAGGCT 5'

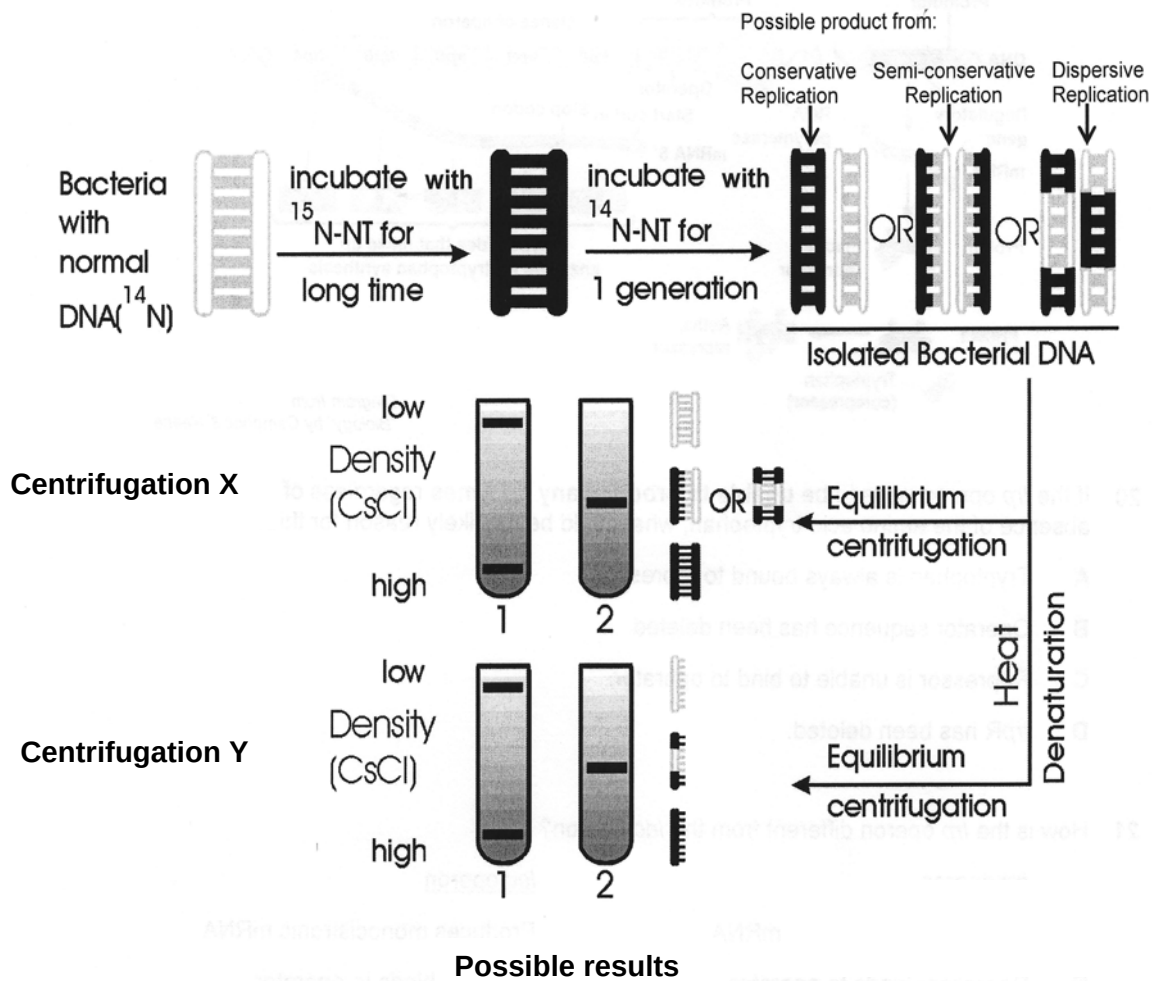
What would be the first four amino acids of a polypeptide generated from this sequence if this gene is expressed and translated?

The mRNA codons for some amino acids are shown in the table below:

Arg	CGA, CGG, AGA, AGG	Leu	CUU, CUC, CUA, CUG
Asp	GAU, GAC	Lys	AAA, AAG
Ile	AUU, AUC, AUA	Phe	UUU, UUC
Start	AUG	Ser	UCA, UCG, AGU, AGC
Stop	UAG, UGA, UAA	Tyr	UAU, UAC

- A** Met-Arg-Ser-Phe
B Met-Arg-Ser-Lys
C Met-Ile-Phe-Leu
D Met-Tyr-Lys-Asp
- 11** If DNA is made up of 2 different types of nitrogenous bases coding for 39 types of amino acids in proteins, what would be the minimum number of bases required for each codon?
- A** 3
B 4
C 5
D 6

- 12 The Meselson-Stahl experiment is shown below. The experiment was done to show that DNA replication occurs by a semi-conservative mode rather than a conservative or dispersive mode.



Given that replication is semi-conservative, which result (tube 1 or 2) would you expect to see for centrifugation X and Y respectively? In addition, which of the other modes of replication would the expected results disprove?

	Centrifugation X		Centrifugation Y	
	Correct result	Disproves	Correct result	Disproves
A	1	Dispersive	1	Conservative
B	2	Conservative	2	Dispersive
C	1	Dispersive	2	Conservative
D	2	Conservative	1	Dispersive

- 13** Listed below are some antibacterial drugs that can affect the synthesis of bacterial proteins.

Anti-microbial drug	Rifampicin	Streptomycin	Tetracycline
Mode of action	Binds to RNA polymerase	mRNA misread during translation	Prevents binding of tRNA to ribosome

Which row shows the correct effects of these drugs on bacterial protein synthesis?

	Rifampicin	Streptomycin	Tetracycline
A	Defective protein synthesised	mRNA does not bind to ribosome	Polymerization of amino acids does not occur
B	mRNA not synthesised	Defective protein synthesised	Polymerization of amino acids does not occur
C	mRNA not synthesised	mRNA does not bind to ribosome	Transcription prevented
D	Transcription prevented	Defective protein synthesised	mRNA does not bind to ribosome

- 14** What do chromosomal aberrations and gene mutations have in common and how are they different?

	Similarity	Difference
A	Both may involve addition of nucleotides.	Gene mutations always produce dominant alleles but not chromosomal aberrations.
B	Both may not result in disorders.	Gene mutations do not involve inversions but inversions of segments of chromosomes do occur.
C	Both affect nucleotide sequences in DNA molecule.	Gene mutations occur within a chromosome but chromosomal aberrations may occur across chromosomes.
D	Both may not result in a difference in protein expression.	Chromosomal aberrations are more harmful than gene mutations.

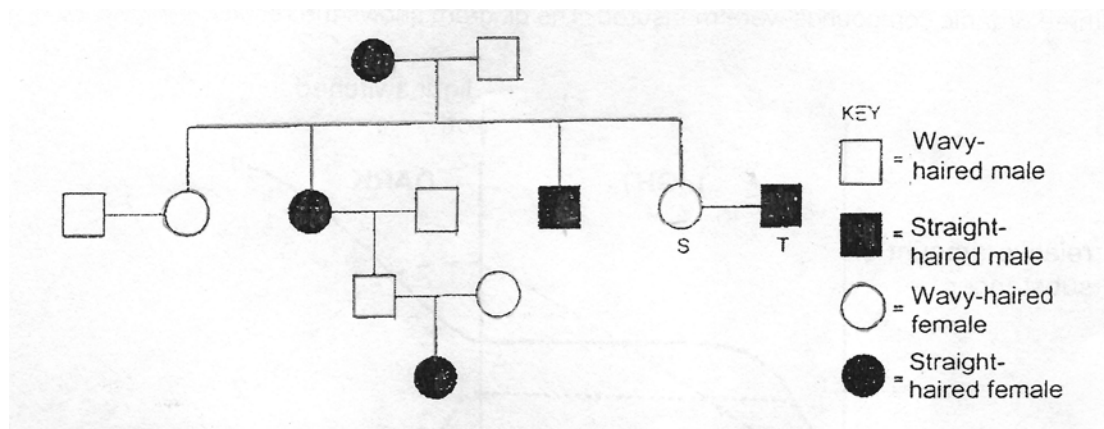
- 15 In pigeons, the alleles of the gene controlling eye colour are codominant and are carried on the X chromosome. When a black-eyed female was mated with a white-eyed male, the F_1 generation consisted of black-eyed males and grey-eyed females.

What phenotypic ratio would be expected in the F_2 generation?

- A 1 black-eyed male: 1 white-eyed male: 2 grey-eyed females
 B 1 black-eyed male: 1 white-eyed male: 1 grey-eyed female: 1 black-eyed female
 C 1 black-eyed male: 1 grey-eyed male: 1 white-eyed female: 1 black-eyed female
 D 2 black-eyed males: 1 grey-eyed female: 1 white-eyed female
- 16 Which one of the following genotypes **cannot** occur amongst the children of a mother with blood group A and a father with blood group B?

- A $I^A I^A$
 B $I^A I^B$
 C $I^A I^O$
 D $I^O I^O$

- 17 The pedigree below shows the inheritance of wavy and straight hair in a family.



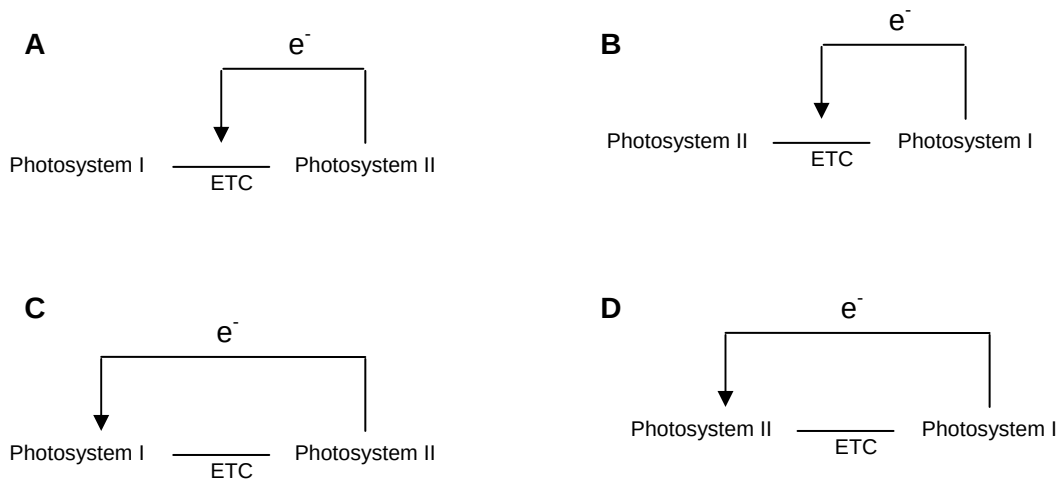
What is the chance of persons **S** and **T** producing a straight-haired girl?

- A 1 in 1
 B 1 in 2
 C 1 in 3
 D 1 in 4

- 18** It is suggested by scientists that prosopagnosia, a condition in which an individual is unable to recognise or distinguish between the faces of other humans, was caused by a single dominant gene. In preliminary studies, up to 2% of tested individuals were affected.

Which of the following would support the hypothesis that it is caused by a single, dominant gene?

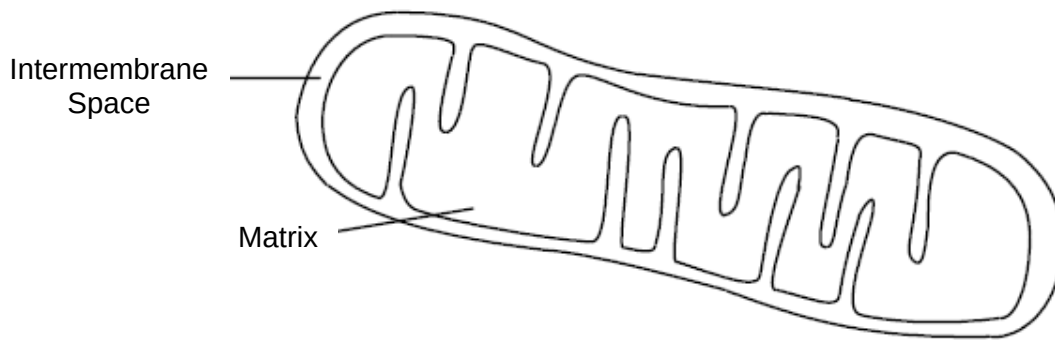
- A** Only males are prosopagnosics.
 - B** As long as one of the parents is a prosopagnosic, all the offspring will be prosopagnosic.
 - C** No offspring of two normal individuals are prosopagnosic.
 - D** The severity of the trait in individuals follows a normal distribution.
- 19** Which diagram correctly represents the process of cyclic photophosphorylation?



- 20** Which of the following statements about the Calvin cycle is **false**?

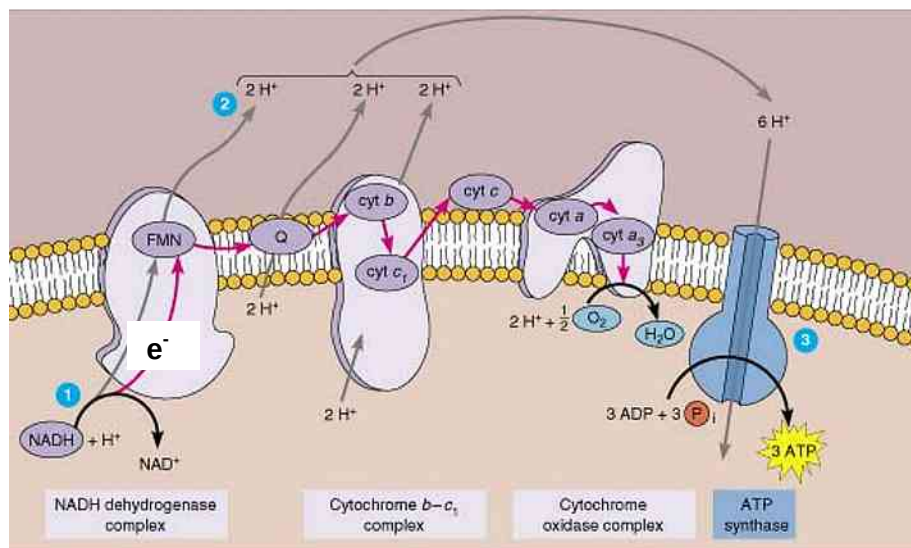
- A** ATP is needed for the regeneration of 5C RuBP from glyceraldehyde-3-phosphate.
- B** One reduced NADP^+ is needed for the reduction of one glyceraldehyde-3-phosphate to one glycerate-3-phosphate.
- C** Rubisco catalyses the formation of an unstable 6C compound from one CO_2 molecule and one RuBP molecule.
- D** Six turns of the Calvin cycle are required for the formation of one molecule of glucose.

- 21 The diagram below shows a mitochondrion as seen under the electron microscope.



What is the advantage of having a small volume inside the intermembrane space of the mitochondrion?

- A A high electron concentration is rapidly developed.
 - B A high proton concentration is rapidly developed.
 - C Enzymes of the Krebs cycle are highly concentrated.
 - D Protein electron carriers are highly concentrated.
- 22 Cyanide poison works by inhibiting cytochrome oxidase, the final member of the electron transport chain in the mitochondrion. The figure below shows the electron transport chain in a mitochondrion.

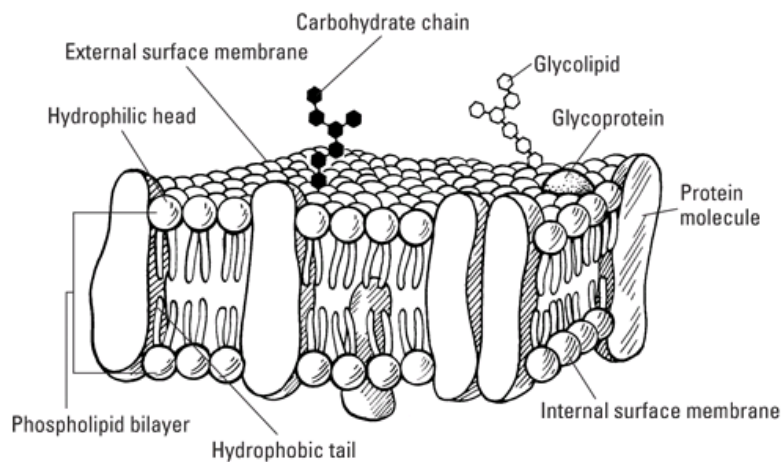


Which of the following will **not** be observed in the mitochondrion when cyanide is introduced?

- A ATP synthesis will stop.
- B Electron flow along the electron transport chain will be interrupted.
- C Formation of pyruvate via glycolysis will continue.
- D No further oxidation of acetyl-CoA.

- 23 Ras protein is coded for by a proto-oncogene, and is an important protein controlling cell division. Which of the following could lead to the cell undergoing uncontrolled cell division?
- A Duplication of the Ras promoter that controls the proto-oncogene.
 - B Loss-of-function mutation to the Ras proto-oncogene.
 - C Point mutation within the proto-oncogene, forming mutant Ras which is constitutively active.
 - D Translocation of the proto-oncogene upstream of a hyperactive promoter.

- 24 The figure below shows a cross-section of a typical cell membrane.



Which of the following statements about the fluid mosaic model of cell surface membranes is **false**?

- A 'Fluid' refers to the phospholipids and proteins being free to move within the membrane.
 - B 'Mosaic' refers to the scattered manner in which membrane proteins are embedded in the cell membrane.
 - C Phospholipids can move laterally and even "flip-flop" within the cell membrane.
 - D Phospholipids and proteins are held together by hydrogen bonds.
- 25 What is the primary reason for natural selection working on every natural population?
- A The genetic variability among organisms of different species.
 - B The innate tendency of organisms in populations to compete against one another.
 - C The need for organisms to fight against natural disasters and diseases.
 - D The tendency for populations to produce more offspring than can be supported.

- 26** Calcitonin is a protein hormone found in humans, fish, birds, and mammals. It is 32 amino acids long. The figure below shows the amino acid sequence of calcitonin in various organisms and the number of amino acid differences when compared with human calcitonin.

Organism	Amino acid sequence of calcitonin	Number of amino acid differences
Human	CGNLSTCMLGTYTQDFNKFHTFPQTAIGVGAP-NH ₂	-
Salmon	CSNLSTCVLGKLSQELHKLQTYPRNTGSGTP-NH ₂	16
Eel	CSNLSTCVLGKLSQELHKLQTYPRTDVGAGTP-NH ₂	16
Rat	CGNLSTCMLGTYTQDLNKFHTFPQTSIGVGAP-NH ₂	2
Chicken	CASLSTCVLGKLSQELHKLQTYPRTDVGAGTP-NH ₂	17

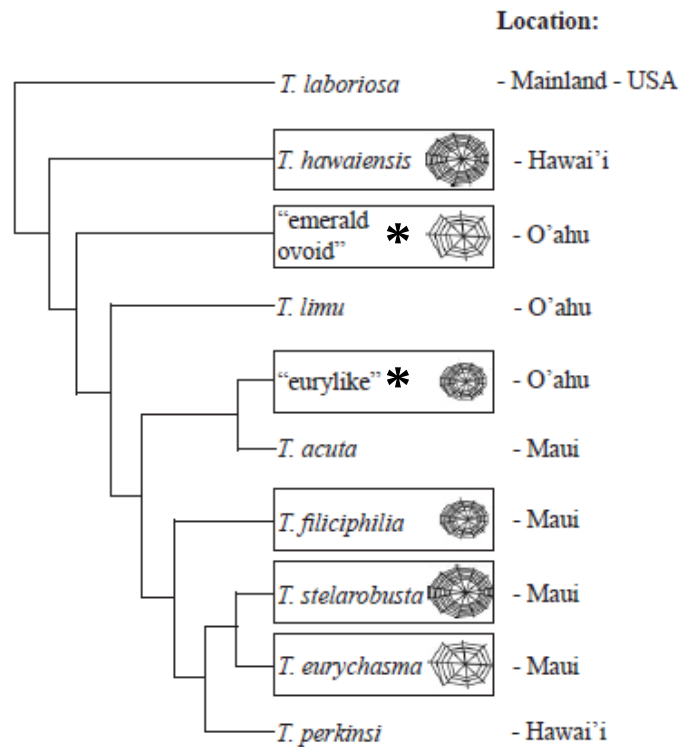
What conclusion can be drawn from the data above?

- A** Humans and salmon are more closely related than salmon and eel.
 - B** Humans are most closely related to rats.
 - C** Salmon and chicken share a recent common ancestor.
 - D** Salmon and eel are more closely related than salmon and chicken.
- 27** Anti-thrombin (a protein involved in blood clotting) and human growth hormone are examples of human proteins that can be produced via the genetic engineering technique.

How would the process which produces anti-thrombin differ from the one that produces bacterial antibiotic resistance protein if both were to be carried out in bacteria?

- A** Antibiotic resistance would not be used as a selection marker for successfully transformed bacteria for the process that produces the antibiotic resistance protein.
- B** The Anti-thrombin gene insert would require an additional eukaryotic promoter.
- C** Anti-thrombin would require additional post-translational modification.
- D** Different plasmids would have to be used.

- 28** The cladogram below shows the classification of a group of spiders found on the Hawaiian islands. An asterisk (*) indicates that this species of spider has yet to be assigned a scientific name.

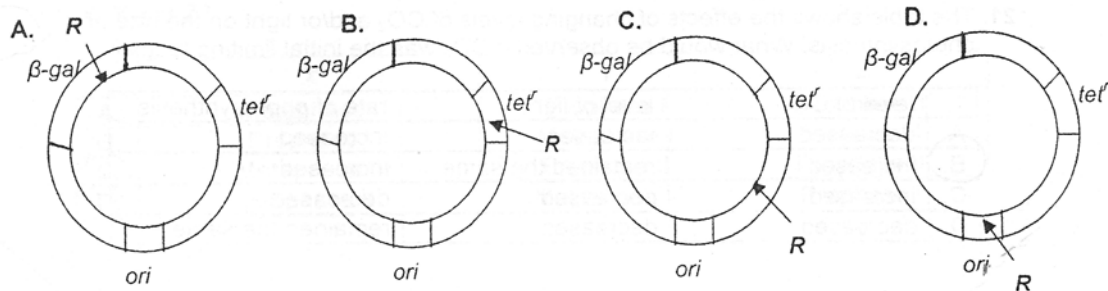


Based on information from the diagram above, deduce which of the following spiders are the most closely related species.

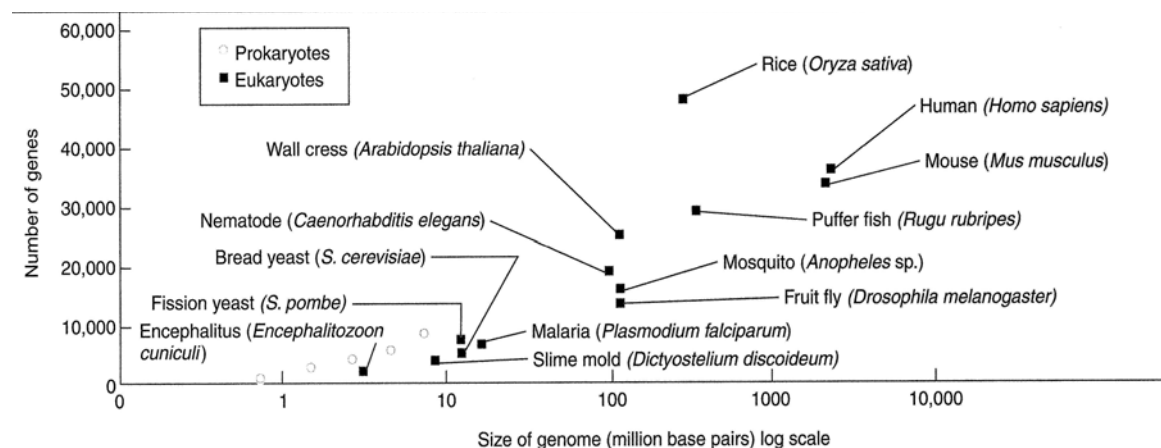
- A** *T. filiciphilia* and "eurylike"
- B** *T. hawaiiensis* and "emerald ovoid"
- C** *T. stelarobusta* and *T. eurychasma*
- D** *T. stelarobusta* and *T. filiciphilia*

- 29 A synthetic plasmid having an origin of replication (*ori*), a restriction site (*R*), two genetic markers, β -galactosidase gene (*β -gal*) and tetracycline-resistance gene (*tet^r*), was used as a vector to transfer the human growth hormone gene into bacteria. The *β -gal* gene codes for an enzyme β -galactosidase, which breaks down a colourless compound called X-gal to a blue compound.

At which position should the restriction site (*R*) be located on the plasmid in order to identify the desired transformed bacteria on an agar medium containing both X-gal and tetracycline?



- 30 The diagram below shows the number of genes and genome size of various types of organisms.



Which of the following conclusions **cannot** be drawn from the diagram above?

- A Generally, eukaryotic genomes are larger and have more genes than prokaryotic genomes.
- B Humans have about twice as many genes as *Drosophila* and fewer genes than rice.
- C The mouse genome is nearly as large as the human genome.
- D The complexity of an organism is reflected in the number of genes it contains.

End of Paper